



FOUR-COURSE SYSTEM

COURSE 3 - DATA ENGINEERING 2026

C3-034C HotelOps Transformation Gate

PROTECTED FRESH FULL RETEST | transformation project under change/recovery

Artifact	C3_034C_HOTELOPS_GATE_RC1
Status	PROTECTED - EXPLICIT ASSIGNMENT ONLY
Phase	C3-02 Transformation Engineering & Warehousing 14 lessons 34-46 focused hours
Prerequisite	C3-01 Gate PASS. C2B warehouse/modeling foundations may be assumed; implementation depth is taught here.
Brand	Charcoal #1F2937 Teal #14B8A6 Soft Blue #3B82F6 AMOLED #000

House rule: analogy -> tiny example -> guided real tool -> break/fix -> independent changed version -> validation -> production reasoning -> evidence. A successful dbt command alone never proves correct data.



Scenario and required architecture

HotelOps receives entity/current-state feeds plus a change delivery. Build the project independently from the supplied synthetic case. Do not copy BrightBasket/practice solution structure blindly; defend every grain and state rule.

Architecture target: supplied source files -> declared raw/source relations -> thin staging -> deterministic current/history/incremental logic -> dimensions/facts/serving model -> tests/reconciliation -> docs/CI/handoff.

REQUIRED BUILD

- Declare raw sources and thin explicit staging models.
- Build a current descriptive dimension with a deterministic surrogate/identity policy.
- Build main fact at declared business grain; add secondary event/detail fact where supplied.
- Create useful serving output without corrupting fact grain.
- Protect grain/required values/relationships with tests plus one custom invariant.
- Apply supplied delta: required update and insert exactly once; rerun same delta unchanged.
- Handle delta-only schema column deliberately and document compatibility/rebuild implications.
- Preserve supplied descriptive change as history while current dimension shows latest truth.
- Reconcile identities/counts and a meaningful amount/duration where present.
- Produce docs/lineage/HANDOFF + clean-run dependency/setup record.
- Run EXPLAIN on one non-trivial query when runtime exists and prove any safe change preserves output.

Competency-only pass rule

PASS requires T1-T7 all PASS and zero critical failures. No weighted-score, average or numeric compensation rule exists.

ID	Competency	PASS evidence	Status
T1	Transformation & staging contracts	Layer boundaries, source/staging contracts and grains are explicit/replayable.	PASS / FAIL
T2	Deterministic models & DAG	Dependencies and current-state selection are deterministic and reviewable.	PASS / FAIL
T3	Testing & integrity	Grain/required/relationship/business tests block true defects; reconciliations prove no silent loss/fan-out.	PASS / FAIL
T4	Incremental & recovery	Update/insert + unchanged rerun safe; rebuild/full-refresh recovery documented.	PASS / FAIL
T5	History	Required descriptive change preserved as legitimate current+historical truth.	PASS / FAIL
T6	Dimensional/late-reference integrity	Facts/dimensions preserve declared grain; late references handled without silent fact loss.	PASS / FAIL
T7	Performance/CI/handoff	Plan reasoning, blocking CI/change safety and reproducible serving handoff are evidenced.	PASS / FAIL



Critical failures

- Wrong fact grain, silent fan-out/loss or materially wrong accepted measure.
- Incremental logic duplicates business identities, misses required update, or same-delta rerun is unsafe.
- Required history overwritten or false versions fabricated.
- A true integrity test is disabled/weakened instead of repaired.
- Late-dimension fact disappears through uncontrolled join/filter.
- Schema/change behavior silently accepted without documented compatibility/rebuild implication.
- Performance change alters business semantics.
- Fabricated dbt/DuckDB/EXPLAIN/test evidence.
- Real secret/private data committed, logged or included in case/evidence package.
- Project cannot be reconstructed from documented source/dependency/setup state.

ATTEMPT FREEZE

Save all Gate work under `evidence/GATE\_C/attempt\_001/`. Freeze file list + hashes (or equivalent immutable snapshot) before opening any Gate review. Later repair/retest evidence is appended; original attempt stays unchanged.

INTERVIEW DEFENSE

Defend architecture, grains, dbt-vs-database boundary, incremental identity, history, one failing test/repair, reconciliation, plan observation, CI/handoff and one tradeoff.